Data Activity 1: Dataset Exploration Summary of Results

Task	Dataset characteristic	Result	Appendix
1.1	Total number of variables (columns)	7	A
1.1	Total number of observations (rows)	270	A
1.1	Variable names	Sno, Date, State.UnionTerritory, ConfirmedIndianNational, ConfirmedForeignNational, Cured, Deaths	A
1.1	Number of numeric variables	5	A
1.1	Number of character/text variables	1	A
1.1	Number of date variables	1	A
1.2	Number of unique states/union territories	27	B
1.2	Earliest recorded date	30 January 2020	B
1.2	Latest recorded date	21 March 2020	B
1.2	Most frequently represented state	Kerala	B

Table 1. Summary of the COVID-19 India dataset's structure and key characteristics.

Appendix A - Task 1.1 R Scripts and R Consol Results

```
> # Upload file
> covid19cases <- read.csv(file.choose())
>
> # Data set structure
> str(covid19cases)
'data.frame':   270 obs. of  7 variables:
 $ Sno                     : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Date                    : chr  "30-01-2020" "31-01-2020" "01-02-2020" "02-02-2020" ...
 $ State.UnionTerritory    : chr  "Kerala" "Kerala" "Kerala" "Kerala" ...
 $ ConfirmedIndianNational : int  1 1 2 3 3 3 3 3 3 3 ...
 $ ConfirmedForeignNational: int  0 0 0 0 0 0 0 0 0 0 ...
 $ Cured                   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Deaths                  : int  0 0 0 0 0 0 0 0 0 0 ...
>
> # Corrects date format as it initially appears as a character format
> covid19cases$Date <- as.Date(covid19cases$Date, format = "%d-%m-%Y")
>
> # Revised Data set structure with date correction
> str(covid19cases)
'data.frame':   270 obs. of  7 variables:
 $ Sno                     : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Date                    : Date, format: "2020-01-30" "2020-01-31" "2020-02-01" "2020-02-02" ...
 $ State.UnionTerritory    : chr  "Kerala" "Kerala" "Kerala" "Kerala" ...
 $ ConfirmedIndianNational : int  1 1 2 3 3 3 3 3 3 3 ...
 $ ConfirmedForeignNational: int  0 0 0 0 0 0 0 0 0 0 ...
 $ Cured                   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Deaths                  : int  0 0 0 0 0 0 0 0 0 0 ...
>
> # Summary statistics
> summary(covid19cases)
      Sno              Date            State.UnionTerritory ConfirmedIndianNational ConfirmedForeignNational     Cured            Deaths
 Min.   :  1.00   Min.   :2020-01-30   Length   :270         Min.   : 0.000          Min.   : 0.00            Min.   :0.0000   Min.   :0.00000
 1st Qu.: 68.25   1st Qu.:2020-03-08   N.unique : 27        1st Qu.: 1.000          1st Qu.: 0.00            1st Qu.:0.0000   1st Qu.:0.00000
 Median :135.50   Median :2020-03-13   N.blank  :  0        Median : 3.000          Median : 0.00            Median :0.0000   Median :0.00000
 Mean   :135.50   Mean   :2020-03-10   Min.nchar:  5        Mean   : 5.552          Mean   : 1.33            Mean   :0.6037   Mean   :0.09259
 3rd Qu.:202.75   3rd Qu.:2020-03-18   Max.nchar: 36        3rd Qu.: 6.000          3rd Qu.: 0.00            3rd Qu.:0.0000   3rd Qu.:0.00000
 Max.   :270.00   Max.   :2020-03-21                        Max.   :60.000          Max.   :14.00            Max.   :9.0000   Max.   :1.00000
>
> # Number of columns/variables
> ncol(covid19cases)
[1] 7
>
> # Number of observations/rows
> nrow(covid19cases)
[1] 270
>
> # Variable names
> names (covid19cases)
[1] "Sno"                     "Date"                    "State.UnionTerritory"    "ConfirmedIndianNational" "ConfirmedForeignNational"
[6] "Cured"                   "Deaths"
>
> # Count of variables in numeric format - specific command, although it can manually be inferred from the Dataset structure command
> sum(sapply(covid19cases, is.numeric))
[1] 5
>
> # Count of variables in character/text format - specific command, although it can manually be inferred from the Dataset structure command
> sum(sapply(covid19cases, is.character))
[1] 1
>
> # Count of variables in date format - specific command, although it can manually be inferred from the Dataset structure command
> sum(sapply(covid19cases, inherits, "Date"))
[1] 1
>
```

Appendix B - Task 1.2 R Scripts and R Consol Results

```
> # Upload file
> covid19cases <- read.csv(file.choose())
>
> # Corrects date format as it initially appears as a character format
> covid19cases$Date <- as.Date(covid19cases$Date, format = "%d-%m-%Y")
>
> # Revised Data set structure with date correction
> str(covid19cases)
'data.frame':   270 obs. of  7 variables:
 $ Sno                  : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Date                 : Date, format: "2020-01-30" "2020-01-31" "2020-02-01" "2020-02-02" ...
 $ State.UnionTerritory : chr  "Kerala" "Kerala" "Kerala" "Kerala" ...
 $ ConfirmedIndianNational : int  1 1 2 3 3 3 3 3 3 3 ...
 $ ConfirmedForeignNational: int  0 0 0 0 0 0 0 0 0 0 ...
 $ Cured                : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Deaths               : int  0 0 0 0 0 0 0 0 0 0 ...
>
> # 1 Showing unique states/territories and frequency
> table(covid19cases$State.UnionTerritory)

                    Andhra Pradesh                          Chandigarh                        Chattisgarh                           Chhattisgarh                               Delhi                             Gujarat
Haryana                   Himachal Pradesh                   Jammu and Kashmir                           Karnataka                                 Kerala                                Ladakh
                                 10                                   1                                   1                                      2                                    20                                   2
18                                    1                                     1                                     13                                     52                                     1
                    Madhya Pradesh                         Maharashtra                             Odisha                            Pondicherry                           Puducherry                             Punjab
Rajasthan                     Tamil Nadu                           Telengana       Union Territory of Chandigarh Union Territory of Jammu and Kashmir          Union Territory of Ladakh
                                  1                                  13                                   6                                      2                                     2                                  13
19                                   15                                  20                                     2                                     12                                    14
                     Uttar Pradesh                         Uttarakhand                          West Bengal
                                 18                                   7                                   4
>
> # 1 Count of how many unique states/territories. Answer = 27
> length(unique(covid19cases$State.UnionTerritory))
[1] 27
>
> # 2 Calculate date range. Answer = 30/01/2020 - 21/03/2020
> range(covid19cases$Date)
[1] "2020-01-30" "2020-03-21"
>
> # 3 Frequency workings. Answer = Kerala
> library(dplyr)
>
> state_counts <- group_by(covid19cases, State.UnionTerritory) %>%
+   summarise(count = n())
> state_counts[order(-state_counts$count), ]
# A tibble: 27 × 2
   State.UnionTerritory      count
   <chr>                     <int>
 1 Kerala                       52
 2 Delhi                        20
 3 Telengana                    20
 4 Rajasthan                    19
 5 Haryana                      18
 6 Uttar Pradesh                18
 7 Tamil Nadu                   15
 8 Union Territory of Ladakh    14
 9 Karnataka                    13
10 Maharashtra                  13
# ℹ 17 more rows
# ℹ Use `print(n = ...)` to see more rows
```